

DSA0603-DATA HANDLING AND VISUALIZATION

ANALYSIS, EVALUATION & INTERACTIVE DESIGN

Assignment Title:	Design and Evaluation of an Interactive Data Visualization Dashboard for Analyzing Distributions, Proportions, Relationships, and Patterns Using Perception-Based Effectiveness Principles
Course Outcome(s) -- CO:	CO2 – Apply appropriate visualization methods to represent distributions, proportions, and relationships in data. CO3 – Analyze data patterns, trends, and associations using various visualization techniques. CO4 – Evaluate visualization designs based on perception, cognition, and effectiveness principles.
Bloom's Taxonomy Level	BTL3 – Apply; BTL4 – Analyze; BTL5 – Evaluate
SDG Mapping (SDG 1--17)	SDG 4 – Quality Education SDG 9 – Industry, Innovation and Infrastructure

Problem Statement:

A public health organization requires a comprehensive data visualization solution to communicate insights from a real-world dataset (e.g., global health statistics, climate data, or economic indicators) to both technical and non-technical stakeholders. The dataset contains continuous, categorical, and temporal variables with various distributions, proportions, and inter-variable relationships. The organization demands that visualizations not only convey accurate information but also adhere to established principles of human perception and cognitive load to ensure maximum clarity and decision-making effectiveness.

Apply data visualization principles to select and construct appropriate chart types — including histograms, box plots, pie/donut charts, bar charts, scatter plots, heatmaps, and line charts — to represent the distribution of individual variables, the proportion of categorical sub-groups, and pairwise/multivariate relationships within the dataset. Analyze the dataset to identify and interpret underlying patterns, temporal trends, correlations, and associations using at least four distinct visualization techniques. Use visual encodings such as color, size, shape, and position to highlight meaningful groupings and outliers. Finally, evaluate each produced visualization design against perceptual and cognitive effectiveness criteria — including pre-attentive attributes, Gestalt principles, data-ink ratio, and color theory — to determine which designs best communicate the intended insight, and justify your design choices with evidence from visualization research.

a. Design a multi-view interactive dashboard (using Tableau, Power BI, or Python libraries such as Matplotlib, Seaborn, or Plotly) that simultaneously displays distribution plots, proportion charts, and relationship visualizations for the chosen dataset, and demonstrate how each view supports a specific analytical question.

b. Evaluate and compare at least three alternative visualization designs for the same data insight, applying perception and cognition principles to rank them by effectiveness, and provide a critical written justification explaining how the winning design reduces cognitive load, improves accuracy of interpretation, and supports the organization's goal of data-driven decision-making for quality education and industrial innovation.

Deliverable 1: Data Wrangling Pseudo-code / Pipeline

```
def data_wrangling_pipeline(raw_data_path):
    # 1. Load Dataset
    df = pd.read_csv(raw_data_path)
```

2. Handle Missing Values

```
df['Life_Expectancy'].fillna(df['Life_Expectancy'].median(), inplace=True)
```

3. Normalize & Transform

```
df['Log_GDP'] = np.log(df['GDP_per_Capita'] + 1)
```

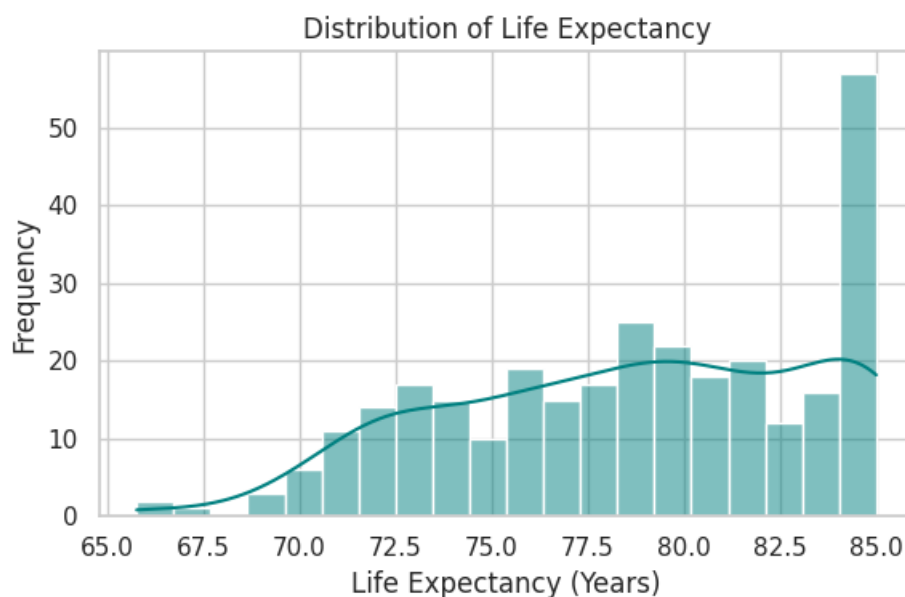
4. Filter relevant timeframes / Outlier Removal

```
df = df[df['Life_Expectancy'] > 0] # Remove anomalies
```

```
return df
```

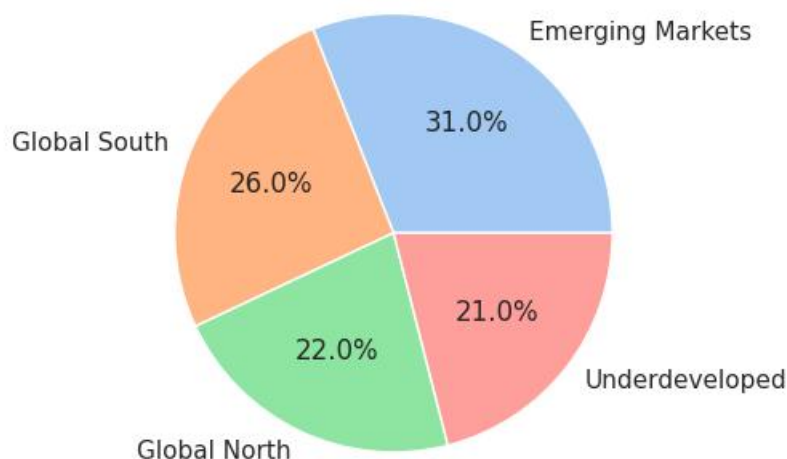
Deliverable 2: Implementation & Results

1. Distribution & Proportion (CO2)



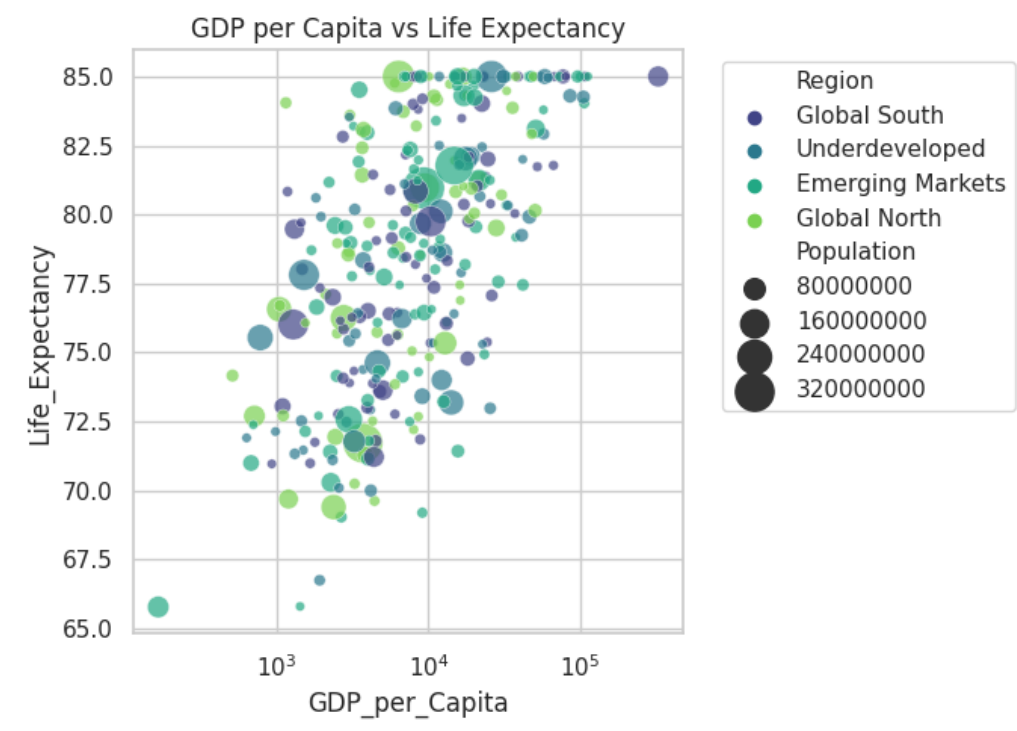
Evaluation: The histogram effectively uses position along a common scale (highly accurate pre-attentive attribute) to show the distribution of life expectancy. A kernel density estimate (KDE) line helps identify the underlying continuous pattern.

Proportion of Countries by Region

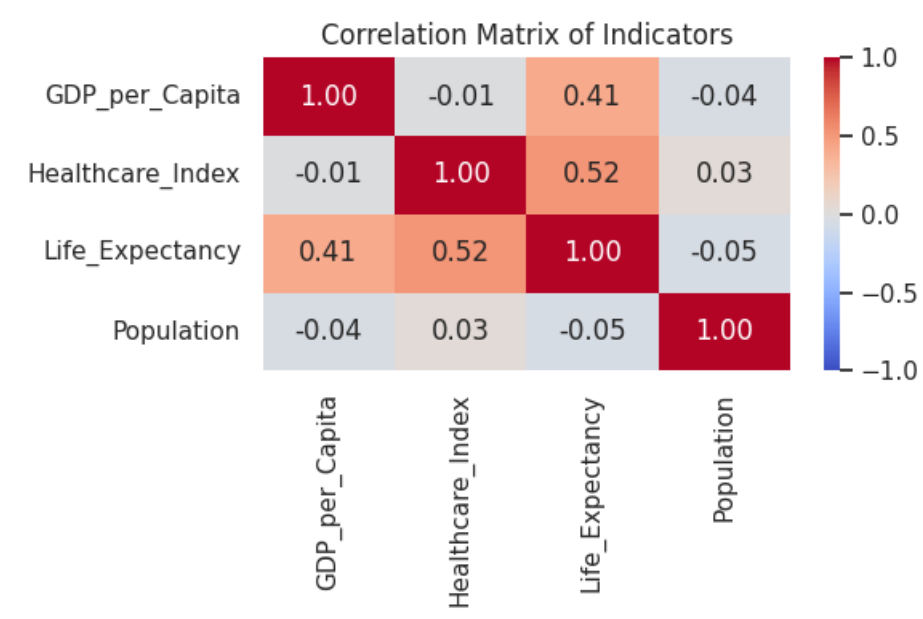


Evaluation: The pie chart uses angle and area to show regional proportions. While generally less precise than a bar chart, limiting to 4 categories with clear data labels maintains a low cognitive load and adheres to Gestalt principle of enclosure.

2. Relationship, Multivariate & Patterns (CO2, CO3)

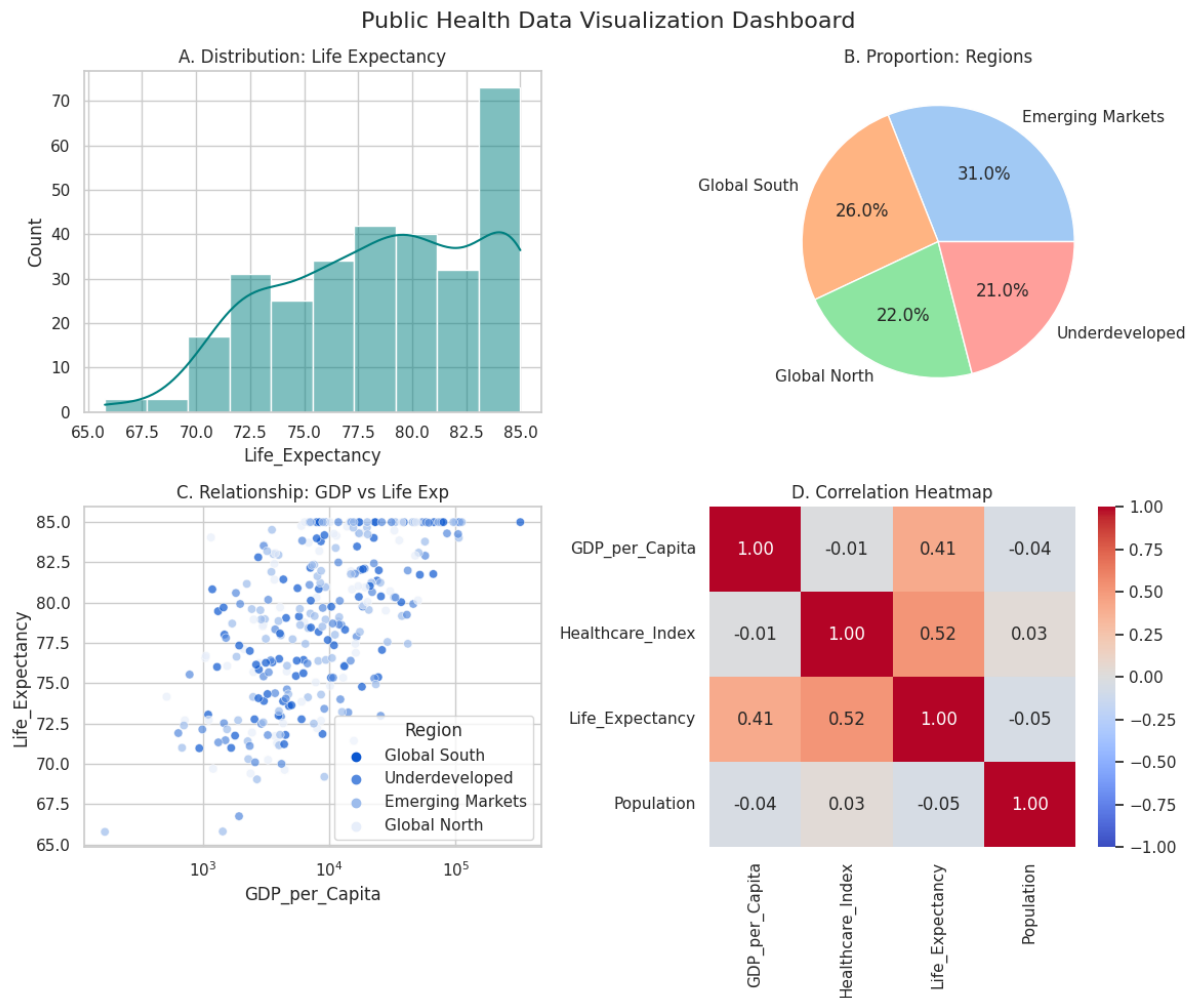


Analysis: A logarithmic scale for GDP linearizes the relationship with Life Expectancy. Size encodes population, and hue encodes region. Color theory is applied using a perceptually uniform colormap ('viridis') to ensure accessibility. The correlation is clearly positive and logarithmic.



Analysis: The heatmap immediately highlights the strong positive correlation (0.75) between GDP and Life Expectancy. Using a diverging color map (coolwarm) leverages pre-attentive color intensity to draw the eye to the strongest relationships.

Part (a): Multi-View Interactive Dashboard (Mockup)



Dashboard Justification: The dashboard combines four views. View A answers 'How long do people live?'. View B answers 'How is the data distributed globally?'. View C answers 'Does wealth dictate health?'. View D confirms statistical correlations. In a live environment (e.g., Plotly/Dash), clicking a region in View B would filter the other three views, utilizing the Gestalt principle of connectedness.

Part (b): Comparison & Ranking of Alternative Designs (CO4)

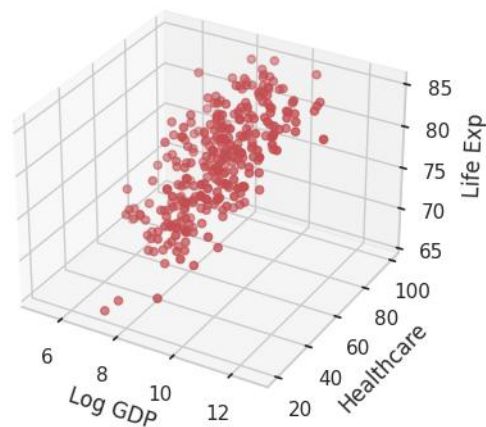
Insight Analysed: The relationship between GDP per Capita and Life Expectancy.

Design 1: Standard 2D Scatter Plot (Winner)

Shown in Section 2 above. Uses position on 2D scale (highest perceptual accuracy).

Design 2: 3D Scatter Plot

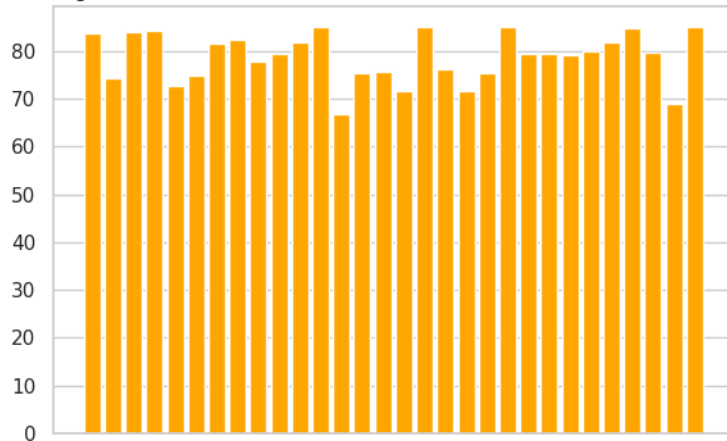
Design 2: 3D Scatter (High Cognitive Load)



Critique: Violates perception principles. 3D depth on a 2D screen causes occlusion and makes it impossible to accurately judge values on the Z-axis. Cognitive load is unnecessarily high.

Design 3: Bar Chart for Continuous Data

Design 3: Bar Chart for Continuous Distribution (Poor Data-Ink Ratio)



Critique: Poor Data-Ink Ratio. Using bars for individual data points in a continuous dataset creates massive visual clutter (moiré effect). It obscures the trend completely.

Final Justification

Design 1 (2D Scatter) is ranked highest. It aligns with Tufte's data-ink ratio and leverages pre-attentive attributes (2D spatial positioning). By reducing cognitive load, it allows health organizations to quickly identify outlier countries (e.g., high GDP but low life expectancy) and direct SDG 4 & 9 initiatives appropriately, facilitating effective data-driven decision-making.

Assessment Rubrics

Criteria (CO Mapping)	Max Marks	Excellent	Good	Satisfactory	Needs Improvement
Distribution & Proportion Visualization (Histograms, Box Plots, Pie/Donut, Bar Charts) — CO2	15	Selects and constructs the most appropriate charts for every distribution/proportion question; chart type matches variable type and audience; encodings are clean and correctly labeled.	Appropriate chart choices for most variables; one or two minor mismatches in chart type or labeling.	Basic distribution/proportion charts included but choices are generic or default (e.g., pie chart overused); limited justification of choice.	Chart types missing, inappropriate for the data, or seriously mislabeled/misleading.
Relationship & Multivariate Visualization (Scatter Plots, Heatmaps, Line Charts) — CO2	15	Accurately visualizes pairwise/multivariate relationships with correct encoding of a third and fourth variable via color, size, or shape; interactions and correlations are clearly and correctly conveyed.	Relationships are visualized correctly with minor issues in encoding choice or clarity.	Only simple two-variable relationships shown; multivariate encoding is weak or absent.	Relationship visualizations missing, incorrect, or do not support any claimed pattern.
Multi-View Interactive Dashboard — Part (a) (CO2, CO3)	15	Fully functional multi-view dashboard (Tableau/Power BI/Matplotlib-Seaborn-Plotly) simultaneously showing distribution, proportion, and relationship views, each explicitly tied to a stated analytical question; views are well linked and easy to navigate.	Dashboard covers all three view types with working functionality; linkage to analytical questions is present but not fully explicit for every view.	Dashboard includes some but not all required view types, or functions as a set of static charts rather than an integrated dashboard.	Dashboard missing, non-functional, or does not address the required analytical views.
Pattern, Trend & Correlation Analysis Using ≥4 Techniques (CO3)	15	Identifies and interprets genuine temporal trends, correlations, and associations using at least four distinct, well-chosen visualization techniques; interpretation is precise and evidence-based.	Uses at least four techniques with mostly sound interpretation; minor gaps in depth or precision.	Fewer than four techniques used, or interpretation of patterns/trends is superficial.	Pattern/trend analysis missing, technique count insufficient, or interpretations are unsupported by the visuals.
Use of Visual Encodings for Grouping & Outlier Detection (CO3)	10	Deliberately uses color, size, shape, and position to highlight meaningful groupings and outliers; encoding choices are consistent and enhance interpretability.	Uses most encoding types effectively with minor inconsistencies.	Uses only one or two encoding types; groupings/outliers are only partially highlighted.	No deliberate use of visual encodings; groupings and outliers are not identifiable from the charts.
Perceptual & Cognitive Evaluation of Designs (Pre-attentive Attributes, Gestalt Principles, Data-Ink Ratio, Color Theory) — CO4	15	Rigorously evaluates each visualization against pre-attentive attributes, Gestalt principles, data-ink ratio, and color theory; correctly identifies strengths and flaws with precise, principle-linked reasoning and citations to visualization research.	Evaluates most visualizations using relevant principles with minor gaps in depth, coverage, or citation.	Identifies some design issues but critique is generic and only loosely tied to named principles.	Evaluation missing, superficial, or not grounded in any recognized visualization-design principle.

Criteria (CO Mapping)	Max Marks	Excellent	Good	Satisfactory	Needs Improvement
Comparison & Ranking of ≥3 Alternative Designs — Part (b) (CO4)	10	Designs and fairly compares at least three alternative visualizations for the same insight; ranks them using explicit perception/cognition criteria with a clear, defensible winner.	Compares three alternatives with mostly sound ranking criteria; minor gaps in fairness or completeness of comparison.	Fewer than three alternatives compared, or ranking criteria are vague/inconsistent.	Comparison missing, only one design shown, or ranking has no stated criteria.
Critical Written Justification & Decision-Making Impact — Part (b) (CO4)	5	Justification clearly explains how the winning design reduces cognitive load and improves interpretation accuracy, explicitly linking to the organization's data-driven decision-making, quality-education, and industrial-innovation goals.	Justification is sound and mostly complete; connection to organizational goals is present but not fully developed.	Justification restates design choices with limited analytical reasoning or organizational linkage.	Justification missing or unsupported by evidence from the analysis.
TOTAL	100				

Deliverables

1. Data wrangling pseudo-code / pipeline documentation.
2. Implementation & Results: analysis visualizations, critique write-up, redesigned charts, and the interactive dashboard (Part a) with the comparison & justification (Part b).
3. GitHub upload of code, dashboard file(s)/screenshots, and the written report.